Final Project

CIS 5450: Big Data Analytics

I. Introduction to the Final Project

The objective of this project is for you to show that you are able to apply the topics in this course in a real-world setting. You are welcome to use concepts beyond the scope of the course in addition to concepts covered in the course; however, a major portion of your project needs to be based on this course's material. You are not allowed to reuse a project from another course. This project is a good opportunity to create something that you could add to your resume, post online in your personal GitHub, or share on LinkedIn.

Projects will be done in **groups of 4 people.** Given the number of people in the class, there will be exactly one group of 5, which will be allowed to form on a first come first serve basis. Make sure all team members are in time zones where you all can work together without problems. If you are unable to find teammates, a good place to start is the Ed Discussion Teammate Search Thread.

II. Role of Project TA

Each group will be assigned to a **project TA.** Your project TA is your mentor for the project and will be the primary person responsible for grading your project (but not the only one). You are **strongly encouraged** to attend your TA's office hours. You should reach out to them directly (private ed post or via email) if you have questions or would like to schedule a time to meet.

Your TA will be assigned **after the proposals are submitted.** You will receive an email from them with feedback.

Note: TAs will be randomly assigned to projects.

Each group must submit a single project proposal (more information below).

III. Project Deadlines & Components Overview

Your final project is expected to have the following components at minimum:

1. Introduction/ Background
2. EDA (Exploratory Data Analysis)
3. Modeling
4. Description of Challenges / Obstacles Faced
5. Potential Next Steps / Future Direction

This project has **4 milestones** in total:

Milestones	**Due Date**
Proposal	March 26th by 11:59 PM EST
Intermediate Check-In Meeting (Virtual)	Between April 13th - 17th
Final deliverables	April 28th by 11:59 PM EST
Presentations	(Tentatively) April 27th-May 3rd

Logistical Notes:

- This is a 4 person group project.
    - **No collaboration outside your group is allowed.**
    - Contribution will be evaluated. See Project Contribution Form**.**
- Again, please use the Ed Discussion Team Search Thread (Spring 2026 Link) to find group members, taking into account potential time zone differences.
- There are **NO late days** for any components of the project. *(We need the full time to grade your project!)*

IV. Project Proposal Guidelines (5 Points)

Due: **Thursday, March 26th by 11:59 PM EST**

Each group is required to submit a **project proposal** through Gradescope by the date indicated above. This is required for all groups.

Your proposal should contain the following information, which is also outlined in the Gradescope Submission Form:

1. All Group Members' Names & Duties:
    a. Breakdown of what each member will be responsible for.
    Note: This is your best method for ensuring everyone in the group will contribute equally! Please know that this is only a rough plan. It is the entire team's responsibility to complete the project on time.
    b. We encourage you to use the teammate search on Ed to form teams. **You need to have your team formed to submit the proposal.**

2. Data Source: Please ensure it meets the dataset size requirements. Also, we would like you to focus on using structured data, and avoid extensive NLP and Computer Vision projects.

    Important note about dataset: Any data used in your project must be free, publicly available, and legal. Failure to comply will lead to an automatic 0 points. Examples of inappropriate data include company data from internship or Kaggle competition data specified for only competition use.

3. Objective and Value Proposition: Explain what you intend to study with your project. What is the ultimate objective? Why is this project interesting?

4. Modeling Plan: What types of models are you considering? Classification vs. Regression? What's your target variable?

5. Hypothesis Testing: Are you considering doing any hypothesis tests? If so, what are your null hypothesis and your test statistic? How do you plan on using topics covered in class (i.e. simulations) to test your hypotheses?

6. Anticipated Obstacles & Challenges: What challenges and obstacles might you anticipate with this project?

While your final deliverable can certainly vary from the proposal that you submit, the proposal component is designed to help make sure that you are on the right track and so you can receive feedback as you go.

IMPORTANT!!!

- Only one student from each group should submit this assignment through Gradescope, but make sure it contains the name of each person in the group.
- When you submit to Gradescope, make sure to use the Gradescope feature to add your group members to the submission as well (this can be done after submitting the assignment) otherwise your teammates will not get a grade in our gradebook!

  ❌**Important note about Gradescope team submission:** Every time you **re-submit**, you need to **re-add** your teammate. Failure to do so will result in a 5 points penalty for the entire team. ❌

V. Intermediate Check-in Meeting (5 Points)

Between **April 13th - April 17th**

During the intermediate check-in week, each team will schedule a **Zoom meeting** with their project TA to discuss the work they have done on the project and work still left to accomplish. This can either be during the TA's Office Hours or during an agreed upon time with all group members and your assigned TA. ***It is your group's responsibility to schedule this in a timely manner, not your Project TA's!***

By this point, we expect you to have completed the **data wrangling and EDA** part of the project as well as have thought about the modeling process by attempting to build your **baseline model**. You should have also begun doing hypothesis tests if you plan on doing any. We will be making sure you are on the right track and answering any questions you may have about your specific project. This meeting shouldn't take more than 20-30 minutes.

❌ Failure to attend the check-in will result in a 5 point penalty for the individual missing the meeting. ❌

VI. Final Deliverables (123 Points)

Due: April 28th by 11:59 PM EST

Each group must submit the following **THREE** deliverables (additional submission details will be posted soon):

1. **Deliverable #1: Annotated Notebook (93 Points)**
   - Every group ***must*** submit ONE complete notebook with all of the code used throughout the project.
   - Think of a notebook similar to the homework assignments. Your notebook would be broken up into separate sections via Markdown sections and should be **very readable.** (Your TA should be able to understand what you are doing in any part).
   - Each section should have an introduction and some description of the findings. You should describe the analysis that you are performing and the results. Interpret any important visualizations and model output.

2. **Deliverable #2: Presentation (20 Points)**

   The purpose of the presentation is to give you a chance to practice communicating your data science project to stakeholders, which is an invaluable skill to have. We can spend hours cleaning the data and fine-tuning the models, but, if potential stakeholders cannot understand the value of the results or use the results to formulate actionable strategies, the modeling effort may not receive the attention it deserves.

   This is an incredibly challenging skill to master. For this project, it may be helpful to think of a potential stakeholder that may find your project interesting or impactful. And, you have 10 mins to pitch the problem statement, method/model, and implications/finding/results to them!

   Every group ***must*** prepare and deliver a recorded 8-10 mins presentation (no more, no less) that provides a summary of your project. Presentations can optionally be delivered live as well. Please include the following details:
   - Objective and value proposition
   - Dataset used
   - Major learnings from EDA (with top 3-5 well-formatted charts that deepened your understanding of the dataset and informed your model)
   - Modeling results (models used, performance, etc)
   - Hypothesis test results (if any)
   - Implications and insights
   - Challenges/limitations/potential future work

Presentation Expectations:

- You should use slides to present the content requested. There are different slide-making applications, feel free to use any that you feel comfortable with.
- You'll be asked to submit your slides as a PDF as well for reference.
- Please do not include code.
- The face of the team member speaking must always be present onscreen. Additionally, only your voices can be used for audio - no text-to-speech software is allowed.

3. **Deliverable #3: Difficulty Concept Location Notebook**

   Every group ***must*** submit ONE complete notebook with detailing where each difficulty concept was used and a brief description as to why.

   Template (please make a copy)

IMPORTANT!!! Like the proposal, only one student from each group should submit this assignment on Gradescope and you should make sure to use the Gradescope feature to add your group members to the submission as well.

❌**Important note about Gradescope team submission:** Every time you **re-submit**, you need to **re-add** your teammates. Failure to do so will result in a 5 points penalty for the entire team.
❌

VII. Project Rubric

A link to the project rubric can be found here.

VIII. Project Contribution (Optional)

All students will have an opportunity to fill out a Google Form at the end of the project to discuss each group member's contributions as well as any additional information we need to know. We encourage you to think carefully about the division of tasks you choose in your project proposal and to use this to help hold your team accountable—each member of the group will receive more or less the same final grade.

X. Dataset Size Requirements

Rows:

A large dataset should have **50,000+ rows**. Please note that this is the threshold for the number of rows which should be a part of the analysis, so rows removed during data cleaning should **not be** counted. You should have 50,000+ rows **after** the data cleaning.

Features:

Additionally, your dataset should have enough interesting **features** to look at apart from the target classification/regression feature. We can not give a number for this as sometimes a few features can hold a rich amount of information and sometimes we need more features to get a complete picture. Generally, you should have **at least 7-10 columns**. But this number is more flexible and can vary widely.

XI. Tips for Success

- Please get started early, and feel free to come to office hours or post on Ed with any questions.
- Once you've locked down a dataset, **DOWNLOAD** it. We had students scraping data from the web in the previous semesters and lost access to the data because the API was updated and no longer compatible with their package.
- Comment and annotate your notebook as you go. This will save you a lot of time in the end.
- Despite the name, Google Colab does not allow simultaneous edits on the same notebook. The best way to approach this is:
    - Try to finalize data cleaning/data wrangling/feature engineering as early as possible, so you all can compile a standardized notebook that is used for modeling.
    - You can copy blocks of codes in Google Colab. (Ask a TA if you don't know how to do this.)

- If your dataset is large, you are encouraged to download your “cleaned data” and import it midpoint to avoid having to re-process the data every time you restart the session.
    - Note: You still need to submit all the codes you used for data gathering (if applicable), data cleaning, data integration, data wrangling, and EDA.
- As you work on the project, be clear on what question you are answering, what you are doing to answer it, and what your takeaways are.
- Email Etiquette Reminder: When communicating with your TA about the project, make sure you cc all your teammates to ensure everyone is on the same page.